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Identifying influential nodes in complex networks based on a hybrid CNN-GNN approach

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Abstract

Identifying important nodes is a critical task in complex networks which has broad applications in the real world. Traditional methods for influential node identification are somewhat limited, as they fail to fully account for the comprehensive structural features of the network. The advancement of deep learning offers a new direction for addressing these challenges. Leveraging the powerful learning capabilities of neural networks, we propose a novel framework called CGCAN, which combines Convolutional Neural Networks (CNN) and Graph Convolutional Networks (GCN) to identify influential nodes in complex networks. In this framework, a neighborhood matrix and feature matrix are constructed for each node, which are then fed into CNN and GCN layers to learn node representations. Finally, the outputs of these modules are combined to compute node importance scores. To validate the effectiveness of the model, experiments were conducted on 20 synthetic networks and 12 real-world networks, using Kendall's tau coefficient and Monotonicity Index as evaluation metrics. The results demonstrate that CGCAN outperforms baseline methods in most of the synthetic and real-world networks.

1 Introduction

Complex systems, ubiquitous in both nature and society, from ecosystems to the internet, are often modeled as complex networks formed by component interactions. The advent of internet technology has further embedded us in a world rich with such networks. A pivotal aspect of complex network analysis is the identification of key nodes, which are instrumental in understanding network structure and dynamics, and in enhancing network robustness and fault tolerance [1-2]. In practical terms, identifying critical nodes has significant implications. For example, in social networks, pinpointing key opinion leaders is vital for gauging social influence and shaping marketing strategies [3]. In transportation networks, detecting congestion points is essential for traffic optimization [4]. Moreover, in epidemiology, understanding disease spread hinges on identifying key nodes for intervention [5].

Conventional methods for assessing node influence focus on structural attributes, such as degree centrality [6] and betweenness centrality [7], which, while useful, are limited by their reliance on network topology alone. Machine learning, however, offers a more nuanced approach by learning from network data to predict node importance. Techniques like HoRW [8], which employ random walks for node representation, have been effective in network classification and importance prediction. Graph embedding is crucial for processing graph-structured data, with deep learning algorithms like CNNs and GNNs at the forefront of influential node ranking due to their advanced representation learning. Recent models, such as RCNN [9] and InfGCN [10], leverage CNNs and GCNs to predict node influence by analyzing neighborhood information and centrality features.

To overcome the limitations of single-metric methods, we propose a new deep learning framework combining CNNs and GCNs with an attention mechanism to consider both local and global topological impacts on node spreading. The paper is organized as follows: Section 2 covers foundational concepts and related work; Section 3 details the CGCAN algorithm; Section 4 presents experimental results; and we conclude with a summary of contributions.

2 Related Work

This section outlines network definitions and reviews key node identification methods: centrality-based, machine learning-based and deep learning-based approaches.

An undirected, unweighted network is defined as $G = (V, E)$ where $|V| = n$ nodes and $|E| = m$ edges. The topology of network can be described by adjacency matrix $A = \{a_{ij}\}_{n\times n}$, where $a_{ij} = 1$ if nodes i and j are connected, and 0 otherwise.

Traditional methods assess node influence using structural and positional information, including local and global topology, and path information. Degree centrality [6] counts a node's neighbors but is limited by its local focus. Enhancements like DC+ [11] and DNC [12] incorporate neighborhood degree and clustering coefficients for a broader assessment. Global methods, such as betweenness [7] and closeness centrality [13], offer precision but at higher computational costs. The K-shell [14] method evaluates a node's core proximity but lacks granularity. Combining local and global information, as

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in Guo et al.'s fusion gravity model, provides a more effective analysis [15].

Machine learning has introduced innovative solutions for identifying critical nodes by learning from network structures and node attributes. Techniques like HoRW [8] use random walks for node representation, aiding in network classification and importance prediction. Yang et al.'s NLC [16] and Lu et al.'s ELSEC [17] combine network embedding with local structure to create centrality metrics, while Du et al.'s VNS [18] employs deep reinforcement learning, leveraging graph embedding and deep Q-networks to enhance accuracy in discovering critical nodes. Wu et al. [19] enhanced node importance evaluation by integrating VGAE-derived low-dimensional features into algorithms, providing a complementary approach to traditional topological methods.

Deep learning has further advanced node identification through graph representation learning, embedding nodes into low-dimensional spaces for efficient analysis. Ou [20] et al. extended RCNN with Multi-Channel RCNN (M-RCNN), incorporating multiple centrality measures (e.g., neighbor degrees, connected communities, and K-core values) into separate input channels, enabling the model to learn the importance of different structural attributes. Ahmad et al. introduced LCNN [21], combining CNNs with local representations like degree and H-index for node ranking.

Graph Neural Networks (GNNs) have become prominent for their ability to capture both local and global network structures. Zhang et al. [22] combined GCNs with CNNs, using a node contraction matrix to extract features and predict influence. Xi et al. developed RSGNN [23], integrating information entropy, node degree, and average neighbor degree to effectively aggregate neighborhood information. Xiong et al. proposed AGNN [24], merging Autoencoders with GNNs to enhance graph embeddings by capturing structural details. Qu et al. created GNR [25], a GNN-based model that calculates node importance using degree, infection score, and dismantling score. Chen et al. [26] proposed DeepELE, combining GNNs with deep reinforcement learning to model information diffusion and identify key nodes.

These methods surpass traditional approaches by capturing complex network relationships and structural properties, offering a robust foundation for influential node identification.

3 Proposed Method

3.1 Model Framework

A node's influence is closely tied to its neighbors. To construct neighbor networks, we use Breadth-First Search (BFS) with a fixed size S. Starting from 1-hop neighbors, we prioritize nodes with higher Degree and K-shell values. If needed, we extend to 2-hop neighbors. The resulting adjacency matrix A represents the neighbor network, with zero-padding for networks smaller than S.

We create a feature matrix F_i of size S×4 using five centrality measures: (1) Degree Centrality * K-shell (DK). (2) Betweenness Centrality (BC). (3) Closeness Centrality (CC). (4) Clustering Coefficient. The selection of these centrality measures is based on their complementary roles in capturing the structural properties of nodes. DK combines local influence (Degree Centrality) and global position (K-shell). The product of these two measures captures both the local influence and global position of a node. BC identifies bridge nodes controlling information flow, CC measures a node's centrality based on average distance to others, and Clustering Coefficient reflects local community connectivity. These centrality measures describe the structural properties of nodes from different perspectives, comprehensively capturing both local and global information. By normalizing these features, we ensure consistent scales across different networks, thereby enhancing the generalization ability of the model.

To better capture the local features and topological structure of a node's neighbors, we employ a Convolutional Neural Network (CNN) to learn from the feature matrix of the neighbor network. Prior to feeding the data into the CNN, we perform a point-wise multiplication between the symmetrically normalized Laplacian matrix $\tilde{A}$ of the neighbor network and the feature matrix F_i. This operation yields a new feature matrix that serves as the input to the CNN module. The point-wise multiplication effectively aggregates the features of adjacent nodes, facilitating the capture of local graph structural information. Moreover, the multiplication with the normalized adjacency matrix contributes to feature smoothing and noise reduction. The CNN module is composed of two convolutional layers, each followed by a BatchNorm layer and an activation layer utilizing the LeakyReLU function to extract local features. Subsequent to the convolution operations, downsampling is conducted through max pooling layers to reduce the dimensionality of the feature maps. Ultimately, the pooled features are flattened and passed through a fully connected layer to produce the output. As the convolutional kernel traverses the feature matrix, it identifies local connectivity patterns between nodes and their neighbors. The application of activation functions and pooling operations within the CNN enables nonlinear transformations and dimensionality reduction of the extracted local information, thereby condensing complex feature matrices into more concise feature vectors.

The global feature learning module combines Graph Convolution and Graph Attention Mechanism to capture the global feature information of nodes in the graph structure. The graph convolution layer is defined as: $H^{l+1} = \sigma(\tilde{A}H^l W^l + b^l)$, where $\tilde{A}$ is the normalized Laplacian matrix, H^l is the feature representation at layer l, W^l is the weight matrix, b^l is the bias, and σ is the activation function. The input feature matrix is first aggregated through the graph convolution layer. The graph attention layer then computes attention weights to capture complex relationships between nodes. This is followed by a second graph convolution layer for further aggregation. Finally, two fully connected layers perform feature transformation and mapping to generate the output feature representation. LeakyReLU activation is applied after each layer to introduce non-linearity. Additionally, Dropout is used to prevent overfitting. By aggregating information from higher-order neighbors, GNNs are capable of capturing the position and influence of nodes within the entire network. They integrate the features of a node with those of its neighbors, thereby generating more comprehensive node representations. Through the iterative aggregation process in multi-layer GNNs, the representations of nodes progressively incorporate more structural information from the network. This enhances the discriminative power among nodes, making it easier to distinguish between important nodes and ordinary ones in the feature space.

To combine the outputs from the CNN and GNN modules, we designed a joint module. This module concatenates the outputs of CNN and GCN along the feature dimension to obtain a combined feature representation. To dynamically adjust the weights of CNN and GCN outputs, we introduce an attention mechanism. Specifically, a linear layer is used to transform the concatenated features, and the attention weights for each node's CNN and GCN outputs are computed using the Softmax function. Finally, these attention weights are used to perform a weighted sum of the CNN and GCN outputs, resulting in the final node feature representation. This fusion process effectively combines local and global information, thereby enhancing the model's ability to identify node importance. The framework is shown in Fig. 1.

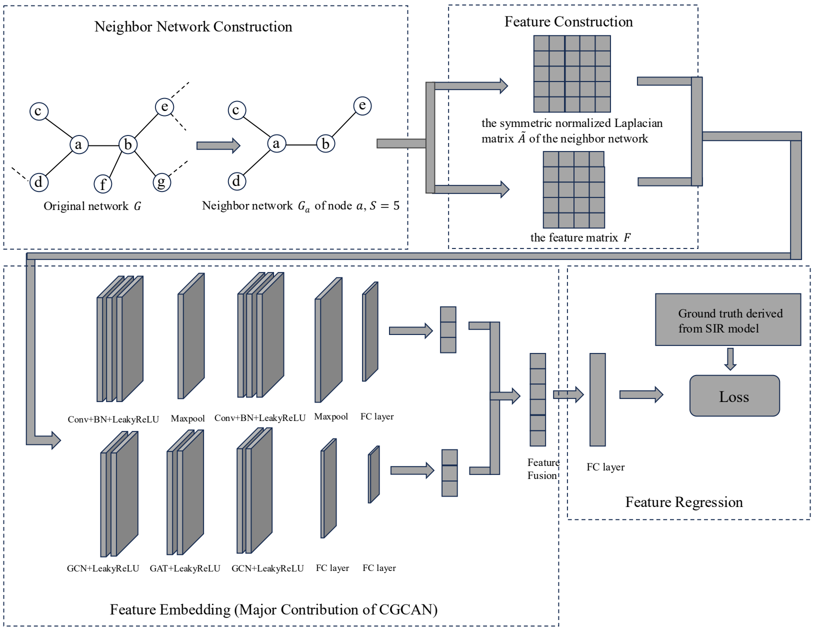

Fig. 1. The CGCAN model framework.

3.2 *Characteristics of CGCAN*

Compared to InfGCN [11], CGCAN introduces several improvements: CGCAN limits neighbor networks to first- and second-hop neighbors, focusing on nodes with significant local influence and reducing computational costs. Instead of BC, we use Degree and K-shell for neighbor ranking, offering lower computational complexity and better adaptability across networks. CGCAN incorporates Degree and K-shell in the feature matrix, providing a balance of local and global structural information to enhance generalization and predictive performance. We integrate CNNs to capture local topology and combines them with GNNs for comprehensive feature learning. The GNN module uses two graph convolution layers and an attention mechanism to model complex graph structures, while the attention mechanism dynamically filters noise, improving overall performance.

4 Experiments

4.1 *Evaluation Metrics*

4.1.1 Kendall's tau Coefficient: Kendall's tau coefficient (τ) measures the similarity between two ranking lists, X and Y. It is defined as: $\tau = \frac{N_c - N_d}{0.5N(N-1)}$, where N is the number of elements, N_c is the number of concordant pairs, and N_d is the number of discordant pairs, τ ranges from [-1, 1], with higher values indicating stronger correlation. We use τ to compare node importance rankings from the SIR model with those from our method and baselines. The SIR model simulates disease spread, categorizing nodes as Susceptible, Infectious, or Recovered. Each infected node infects its neighbors with probability β and recovers with probability γ (set to 1). Node labels are generated by averaging results from 1,000 independent simulations. The infection threshold β_θ is calculated as: $\beta_\theta = \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle}$, where k is node degree, and $\langle \cdot \rangle$ denotes the average calculation.

4.1.2 Monotonicity Index: The Monotonicity Index (MI) measures ranking consistency and distinguishability. It is defined as: $MI = \left(1 - \frac{\sum_{r \in R} n_r(n_r - 1)}{n(n-1)}\right)^2$, where n is the number of nodes, R is the ranking result, and n_r is the number of nodes with rank r. *MI*=0 indicates no distinguishability, while *MI*=1 indicates perfect monotonicity.

4.2 *Datasets*

To validate the effectiveness and discriminative capability of the CGCAN model, this paper conducts simulation experiments, including training on 20 synthetic networks and testing on 12 real-world networks. The statistical properties of the real-world networks are presented in Table 1, encompassing social networks, biological networks, and others.

BA network. Generated by the BA model, with varying node counts N and average degrees $\langle k \rangle$. The Barabási-Albert (BA) model creates scale-free networks characterized by a power-law degree distribution.

Zebra. A network representing contact relationships among 28 Grevy's zebras in Kenya, where nodes denote zebras and edges indicate their contact during the study period.

HIV. A sexual contact network recording the early spread of Human Immunodeficiency Virus (HIV) in the United States

Political Books. A network of books about U.S. politics published during the 2004 U.S. presidential election and sold by Amazon. Edges represent frequent co-purchasing of books by the same buyers.

Jazz. A collaboration network among jazz musicians, where nodes represent musicians and edges denote co-performances in bands.

Wikibooks Edit (na). A Nauru Wikibooks network consisting of users and pages connected by editing events, with each edge representing an edit operation.

Email. An email communication network at Rovira Virgili University, where nodes represent users and edges indicate at least one email exchange.

Wiktionary Edit (na). A Nauru Wiktionary network consisting of users and pages connected by editing events, with each edge representing an edit operation.

Caenorhabditis Elegans. A metabolic network of the nematode Caenorhabditis elegans, where nodes represent substrates and edges denote metabolic reactions.

Figeys. A human protein-protein interaction network derived from large-scale mass spectrometry studies.

Facebook. An ego network from Facebook, including the ego node and all its friends.

Power Grid: The network of the Western U.S. power grid, where nodes represent generators, transformers, or substations, and edges denote power supply lines.

Route Views. An undirected network of interconnected Internet autonomous systems (AS), where nodes represent AS and edges indicate communication between systems.

Table 1. Statistical properties of real-world network datasets used.

Network	n	m	$<k>$	k_{max}	NCC	LCC	c	d
Zebra	27	111	8.2222	14	2	23	0.8759	0.3162
HIV	40	41	2.0500	8	1	40	0.0417	0.0526
Political	105	441	8.4000	25	1	105	0.4875	0.0808
Jazz	198	2742	27.6970	100	1	198	0.6175	0.1406
Wikibooks(na)	250	279	2.2320	77	1	250	0.0274	0.0090
Caenorhabditis	453	2025	8.9404	237	1	453	0.6465	0.0198
Email	1133	5451	9.6222	71	1	1133	0.2202	0.0085
Wiktionary(na)	1470	3300	4.4898	625	2	1468	0.0007	0.0031
Figeys	2239	6432	5.7454	314	9	2217	0.0400	0.0026
Facebook	4039	88234	43.6910	1045	1	4039	0.6055	0.0108
Power Grid	4941	6594	2.6691	19	1	4941	0.0801	0.0005
Route Views	6474	13859	4.2926	1460	1	6474	0.2522	0.0007

4.3 Experimental Results Analysis

The loss function of the model is chosen as Mean Squared Error (MSE), and the activation function employed is LeakyReLU. All learnable parameters are optimized using the Adam optimizer, with the learning rate set to 0.0001 and the training epochs fixed at 200 and a batch size of 32. Early stopping was applied to prevent overfitting. The model is trained on BA synthetic networks with node counts of 500, 1000, 2000, and 3000, and average degrees of 2, 4, 6, 8, and 10. The size of the neighbor network is fixed at $S = 28$. Labels are generated using the SIR propagation model, with the infection rate set to $1.5\times\beta_\theta$, where β_θ represents the propagation threshold of the network. To comprehensively evaluate the model's performance under different infection probabilities and network structures, and to observe the impact of the infection rate β on the experimental results, multiple experiments are conducted with β/β_θ ranging from 1.0 to 2.0. The effectiveness of each method is assessed by calculating the Kendall correlation coefficient and monotonicity index between the experimental results and the outcomes of the SIR propagation model. A higher Kendall coefficient indicates that the ranking of node influence is closer to its actual influence, while a higher monotonicity index suggests a stronger ability of the method to distinguish node influence.

Simulations were conducted on 12 real-world networks, and the results are shown in Fig. 2. The CGCAN model demonstrated strong performance across all networks, with Kendall tau values significantly outperforming baseline methods in some cases. While baseline methods like DC and K-shell excelled in specific networks (e.g., Political Books), and CC and InfGCN performed well in others (e.g., Figeys), their performance was inconsistent across different networks. In contrast, CGCAN consistently achieved high correlation coefficients across all 12 networks, showcasing its general applicability. By capturing multiple node features and leveraging CNNs and GCNs to learn local and global topological information, CGCAN outperformed single-metric centrality measures and single-network-based methods.

The Monotonicity Index results, presented in Table 2, further highlight the superiority of deep learning-based methods. For example, in the Route Views network, CGCAN achieved an MI of 0.9999, compared to 0.9950 for InfGCN and 0.9933 for CGNN, while traditional methods like DC and BC achieved significantly lower MI values (0.5012 and 0.4577, respectively). Additionally, as shown in Fig. 2, CGCAN outperformed CGNN, InfGCN, and RCNN in terms of Kendall tau, demonstrating its superior ability to rank node importance with high distinguishability and correlation.

Table 2. The Monotonicity Index of various methods across 9 real-world networks.

Network	DC	BC	CC	KS	InfGCN	CGNN	RCNN	CGCAN
Zebra	0.7410	0.7630	0.7357	0.3376	0.9225	0.9531	0.2378	**0.9738**
HIV	0.4669	0.4954	0.9369	0.1479	**0.9872**	0.9695	0.3646	0.9847
Political	0.8252	0.9974	0.9847	0.4949	1.0000	1.0000	0.7470	**1.0000**
Jazz	0.9659	0.9885	0.9878	0.7944	0.9995	0.9993	0.9780	**0.9999**
Wikibooks(na)	0.2039	0.1999	0.7913	0.0754	0.9098	0.8043	0.0253	**0.9165**
Caenorhabditis	0.7922	0.8741	0. 9900	0.6962	0.9972	0.9967	0.9243	**0.9993**
Email	0.8874	0.9400	0.9988	0.8088	0.9999	0.9990	0.1855	**0.9999**
Wiktionary(na)	0.5699	0.6813	0.9243	0.5315	**0.9613**	0.9412	0.3373	0.9414
Figeys	0.5970	0.6279	0.9933	0.5677	0.9945	0.9936	0.3165	**0.9982**
Facebook	0.9739	0.9855	0.9967	0.9419	0.9998	0.9999	0.9203	**0.9999**
Power Grid	0.5927	0.8320	0.9998	0.2460	0.9999	0.9963	0.1376	**0.9999**
Route Views	0.5012	0.4577	0.9931	0.4388	0.9950	0.9933	0.8267	**0.9999**

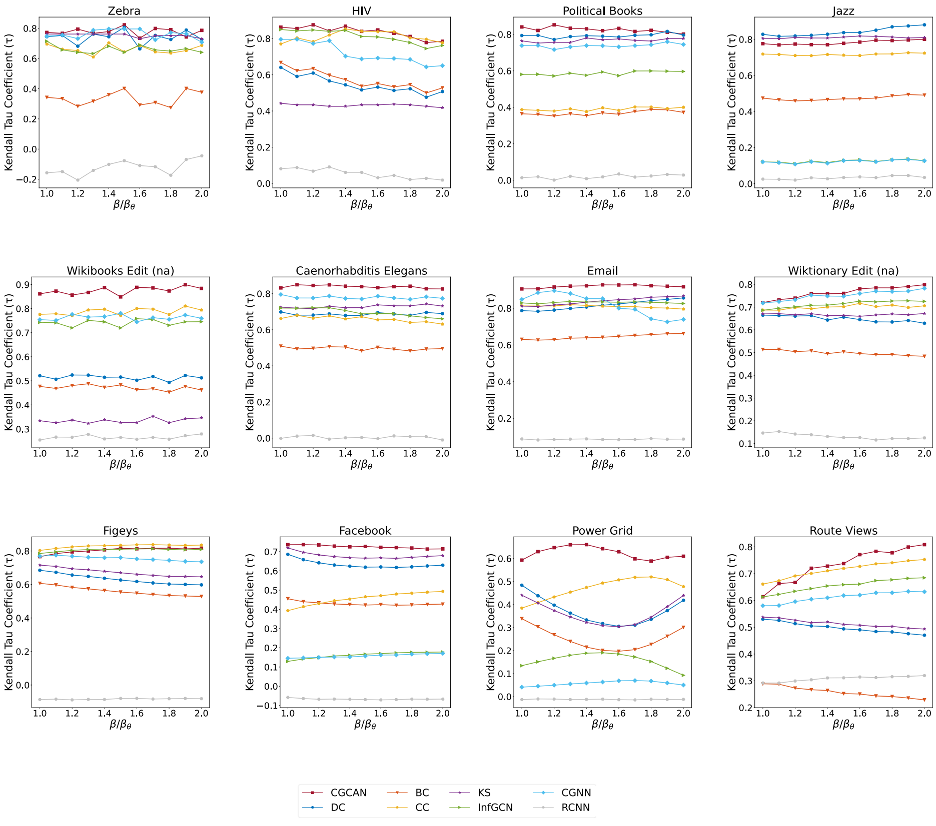

Fig. 2. The Kendall tau correlation coefficient across 12 real-world networks.

5 Conclusion

Identifying critical nodes in complex networks is crucial for understanding network structure, optimizing performance, and enhancing robustness. Traditional centrality measures often focus narrowly on node features or network structure, limiting their effectiveness. Deep learning offers innovative solutions to these challenges. We propose a novel framework combining Convolutional Neural Networks (CNNs) and Graph Convolutional Networks (GCNs) to identify influential nodes. The framework constructs fixed-size neighborhood networks using the BFS algorithm and generates feature matrices based on node characteristics. By integrating CNNs, GCNs, and an attention mechanism, the model captures both local and global topological influences on node importance. Trained and tested using SIR model-generated labels, our method outperforms baseline approaches, demonstrating superior distinguishability and effectiveness in identifying critical nodes.

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